

ScrollAnchor

Exploratory 2D Render-Anomaly Review

PHercParis4 segment w110-112

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Repository: <https://github.com/olgaiv39/scroll-anchor>

Run date: 2026-07-18

Source render:

PHercParis4-20260623163339-2.4um-0.22m-78keV-volume-20260411134726-alpha-overlay-combined-ds8.jpg

This report summarizes an exploratory, read-only 2D analysis of a single downsampled surface render (a JPG). The analysis flags candidate visual discontinuities that may correspond to sheet skips or local render shifts. It does not use surface-normal geometry or CT voxel evidence, so the candidates listed here are starting points for manual review, not confirmed segmentation errors.

Exploratory review candidates - not confirmed sheet skips

Run summary

JPG dimensions (row x col)	6270 x 24030 px
Processed dimensions (row x col)	3135 x 12015 px
Working downsample	2 (half linear resolution)
Whole-render coverage (mapped x8)	50160 x 192240 px
Runtime	318.727 s
Peak memory (RSS)	2967.6 MB
Candidates exported for review	200

The complete downsampled JPG was scanned at half its linear resolution. This is a whole-render pass, not a partial crop.

Source render derived from Vesuvius Challenge open data (PHercParis4 segment 20260623163339-w110-112). The source render is not included in this repository, and this report does not imply endorsement by the Vesuvius Challenge.

Method and candidate counts

The detector scans the render for short seams where the texture on one side aligns with the other only after a small lateral shift. It combines a fine and a coarse scale, weights each location by local texture reliability, and pools evidence along candidate seams. Connected pixels above a fixed pixel-level anomaly threshold form raw components; small components are removed by a minimum-size filter; the remaining components are ranked by a region score and the top-ranked subset is exported for review.

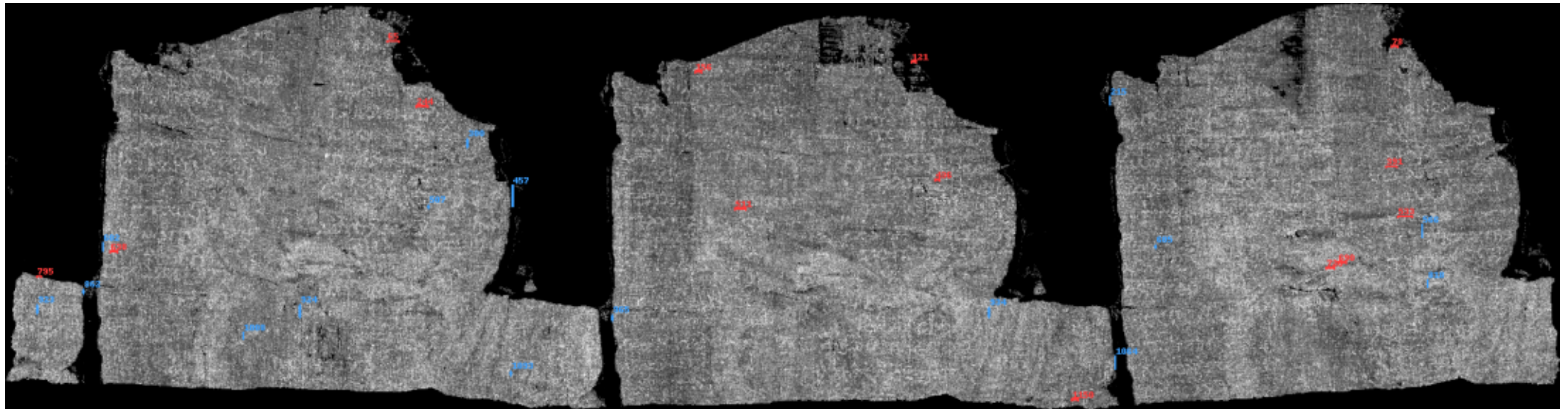
Candidate count funnel

Raw components above pixel anomaly threshold	1152
Components passing minimum size	428
Ranked candidates exported for review	200
Additional passing candidates not exported due to cap	228
Export cap (max_regions)	200

Pixel threshold versus region score

Two different numbers control this pipeline. The pixel anomaly threshold (`anomaly_thr = 0.15`) is applied to the per-pixel anomaly map to decide which pixels are candidates, and a minimum size of 40 processed pixels removes tiny components. Each surviving component then receives a region score, computed as its mean pixel anomaly multiplied by a texture-reliability weight of at most one. A region score can therefore fall below 0.15 even though its pixels passed the pixel threshold. The exported score range here is 0.133 to 0.201. The exported regions are a ranked subset for review, not a count of confirmed errors, and the raw component total is not a count of sheet skips.

Full-render overview



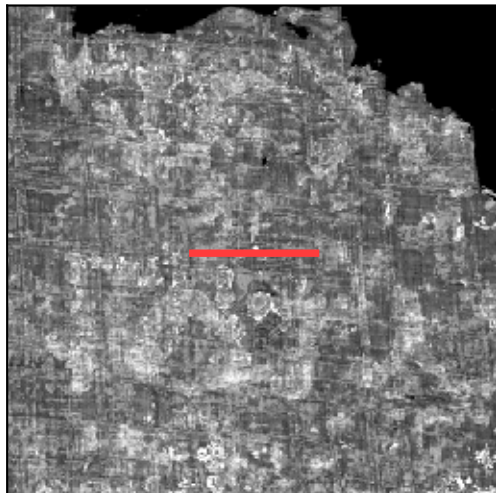
Numbered boxes mark the top-ranked exported candidates over the whole downsampled render.

Legend: red = horizontal seam candidate blue = vertical seam candidate

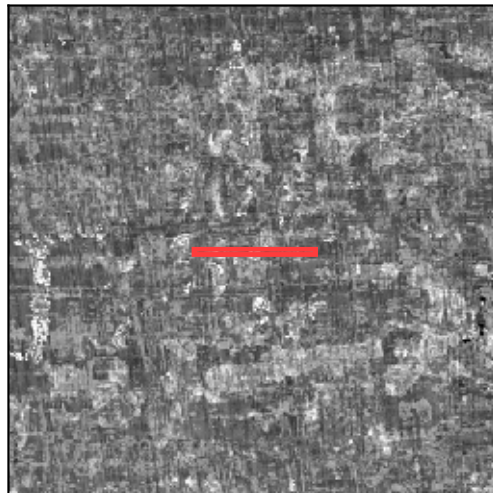
ScrollAnchor - <https://github.com/olgaiv39/scroll-anchor>

Enlarged candidate crops (1 of 2)

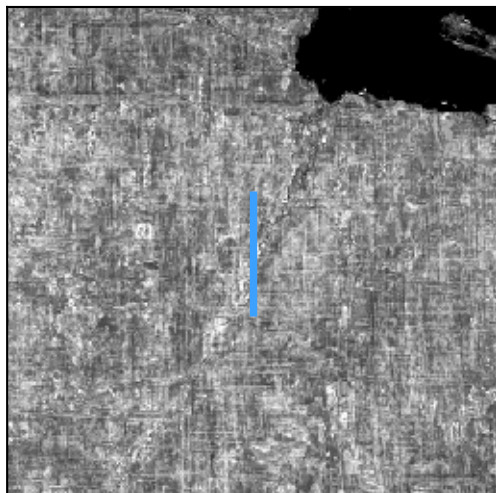
#1 id 244 score 0.201
horizontal 2D shift 14 px
JPG row 1593, col 6438



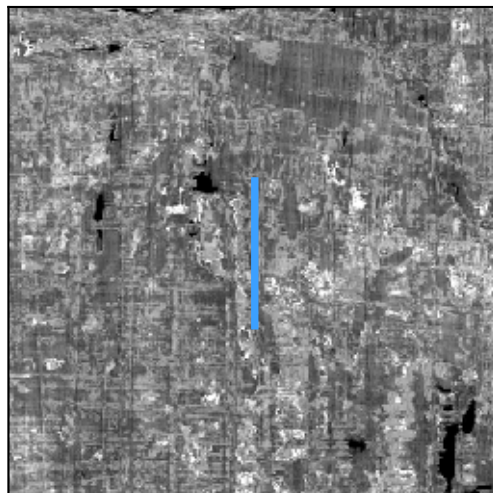
#2 id 391 score 0.186
horizontal 2D shift 6 px
JPG row 2519, col 21432



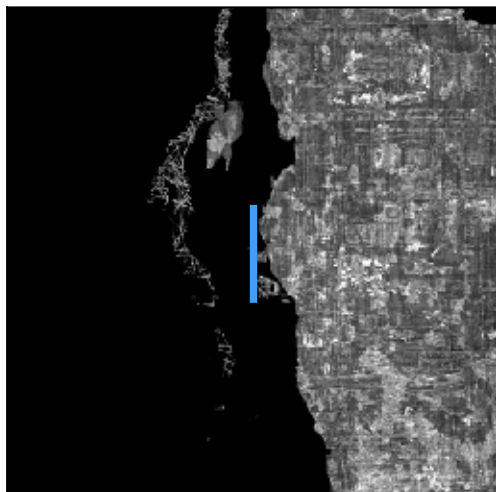
#3 id 934 score 0.185
vertical 2D shift 10 px
JPG row 4775, col 15214



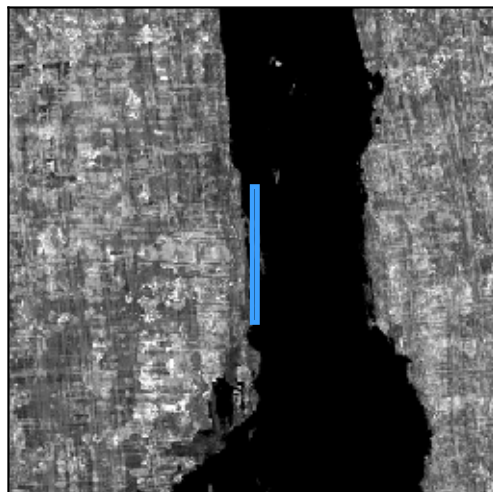
#4 id 924 score 0.183
vertical 2D shift n/a
JPG row 4745, col 4552



#5 id 215 score 0.182
vertical 2D shift n/a
JPG row 1509, col 17079



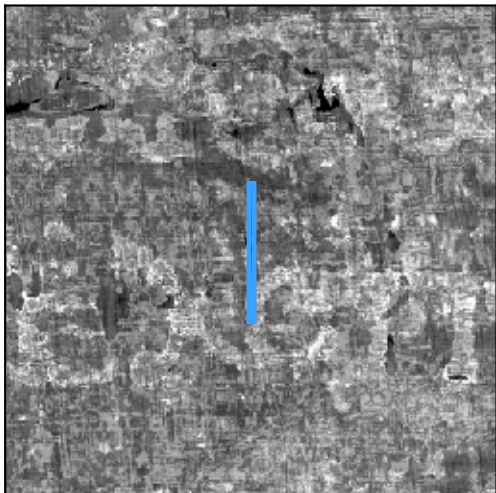
#6 id 1064 score 0.181
vertical 2D shift 14 px
JPG row 5540, col 17163



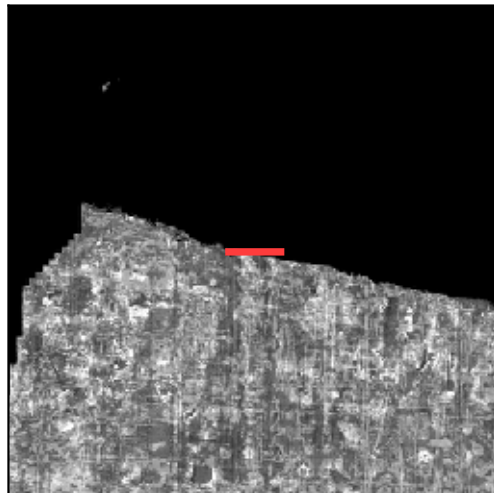
Crops are local context windows on the processed render. A genuine candidate shows a texture offset across the marked box.

Enlarged candidate crops (2 of 2)

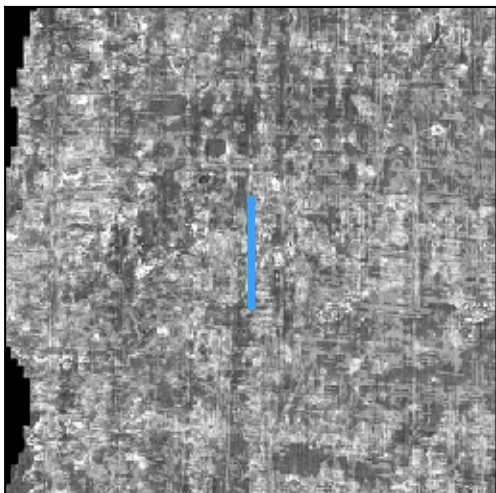
#7 id 566 score 0.178
vertical 2D shift 12 px
JPG row 3509, col 21907



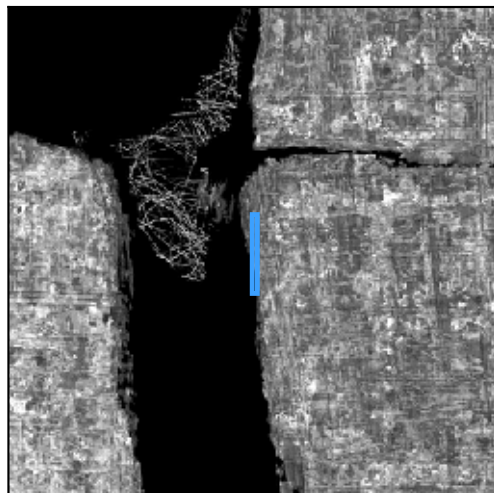
#8 id 795 score 0.172
horizontal 2D shift 14 px
JPG row 4212, col 516



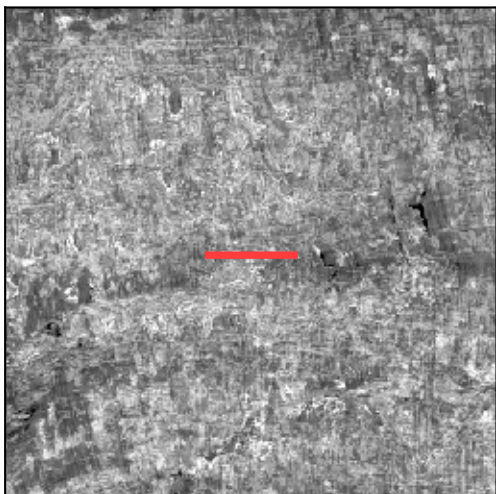
#9 id 923 score 0.171
vertical 2D shift 16 px
JPG row 4715, col 496



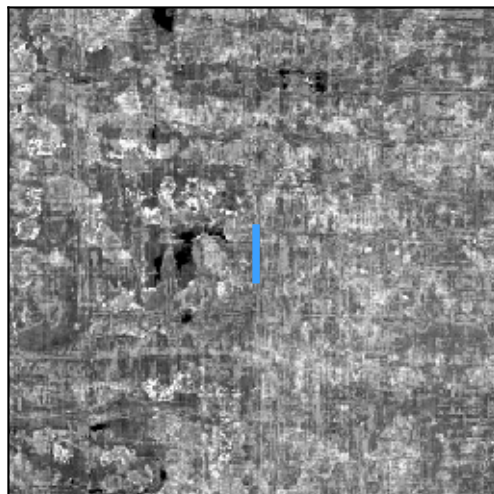
#10 id 965 score 0.170
vertical 2D shift 16 px
JPG row 4860, col 9383



#11 id 690 score 0.169
horizontal 2D shift 16 px
JPG row 3999, col 20665



#12 id 507 score 0.169
vertical 2D shift n/a
JPG row 3140, col 6550



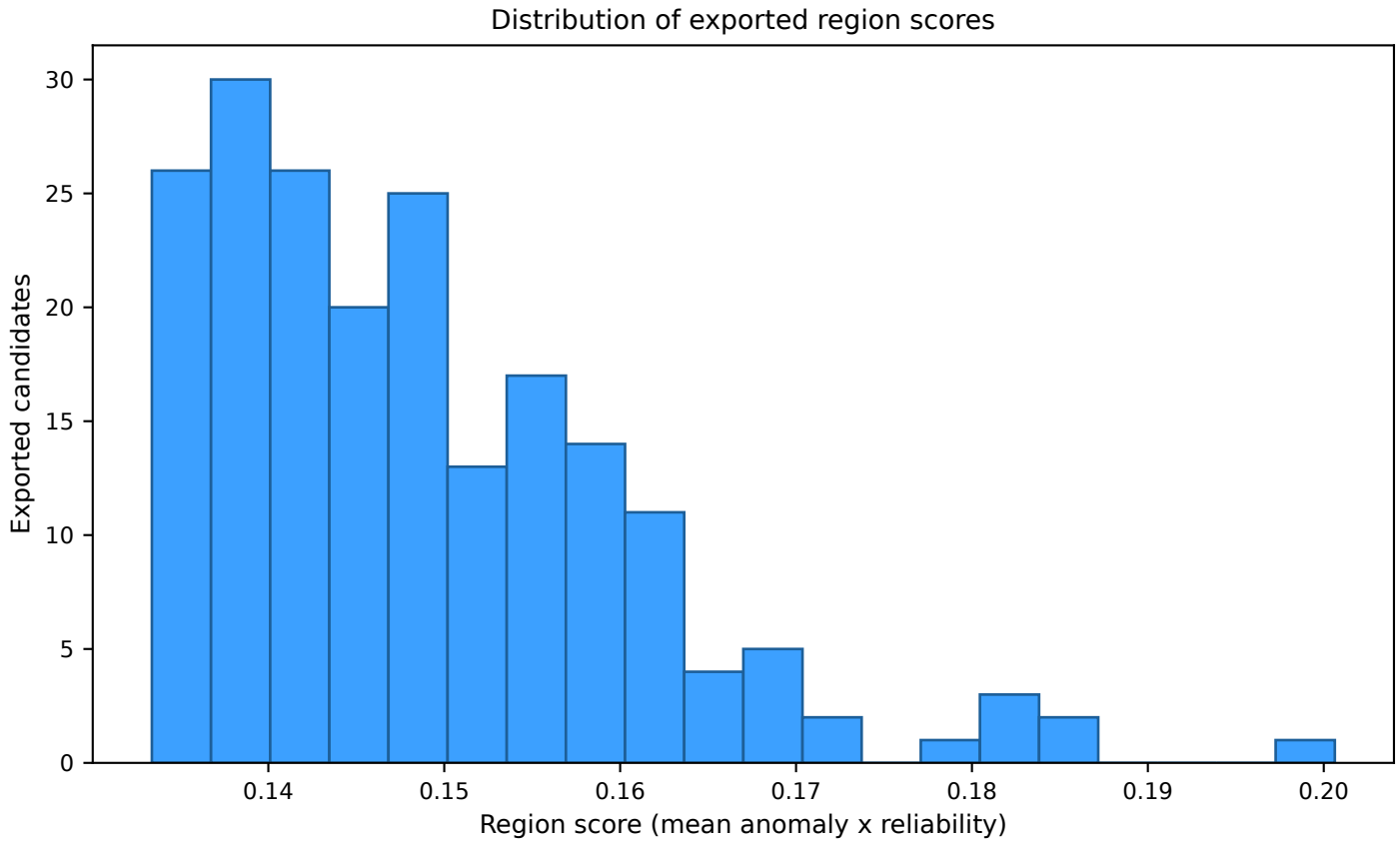
Crops are local context windows on the processed render. A genuine candidate shows a texture offset across the marked box.

Ranked candidates (top 20)

Rank	ID	Score	Direction	Size(px)	JPG row	JPG col	Full row	Full col	2D shift(px)
1	244	0.201	horizontal	155	1593	6438	12742	51506	14
2	391	0.186	horizontal	253	2519	21432	20156	171453	6
3	934	0.185	vertical	243	4775	15214	38198	121713	10
4	924	0.183	vertical	280	4745	4552	37956	36419	n/a
5	215	0.182	vertical	155	1509	17079	12073	136635	n/a
6	1064	0.181	vertical	384	5540	17163	44318	137303	14
7	566	0.178	vertical	333	3509	21907	28071	175255	12
8	795	0.172	horizontal	105	4212	516	33694	4125	14
9	923	0.171	vertical	198	4715	496	37721	3966	16
10	965	0.170	vertical	208	4860	9383	38876	75066	16
11	690	0.169	horizontal	130	3999	20665	31992	165320	16
12	507	0.169	vertical	43	3140	6550	25120	52400	n/a
13	605	0.168	vertical	43	3754	17782	30032	142256	n/a
14	533	0.168	horizontal	565	3285	21644	26278	173155	8
15	638	0.167	horizontal	132	3833	1679	30664	13432	16
16	78	0.166	horizontal	107	669	21486	5351	171890	16
17	1150	0.166	horizontal	57	6106	16536	48848	132288	16
18	457	0.165	vertical	1589	2980	7842	23841	62734	n/a
19	300	0.163	vertical	173	2166	7145	17329	57161	n/a
20	511	0.162	horizontal	93	3166	11360	25328	90883	6

Coordinates are JPG-pixel centroids and their mapped full-render positions. The 2D shift is a local estimate in JPG pixels, not a 3D voxel offset.

Score overview (exported candidates only)



Direction and displacement

Horizontal seam candidates	74
Vertical seam candidates	126
Candidates with a 2D shift estimate	161
Candidates without a 2D shift estimate	39

Scores cluster just above the export floor, which is expected for a conservative exploratory detector. A displacement is only reported when a region's per-pixel alignment is consistent enough to estimate one.

Reviewer checklist and limitations

Suggested review steps

1. Open the full-render overview and locate the numbered boxes.
2. For each candidate, inspect its crop for a genuine texture offset across the box versus an illumination or JPEG compression artifact.
3. Cross-check promising candidates against the 3D segmentation in VC3D using the mapped full-render coordinates in the ranked table.
4. Treat the ranking as a guide only. Low-scoring candidates near the export floor are frequently texture effects rather than sheet skips.

Limitations

Render-only 2D analysis. No surface-normal geometry or CT evidence. Candidates are exploratory visual anomalies on a render for manual review, not confirmed sheet switches, 3D drift, or voxel displacement. 2D displacement is in render (JPG) pixels only. Full-render coordinates are a documented x8 mapping, not verified VC3D coordinates.

This is a render-only 2D screening aid. It carries no surface-normal geometry and no CT evidence, and it makes no claim about 3D voxel displacement or corrected surface coordinates.